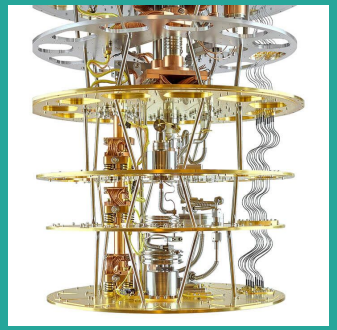
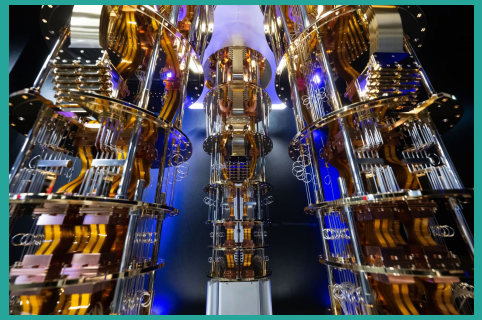
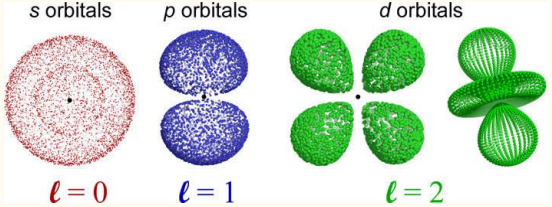




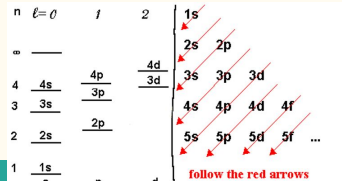
Quantum Computing Playground

By: Uday Goel, Pitt 2030 (QC + Physics)

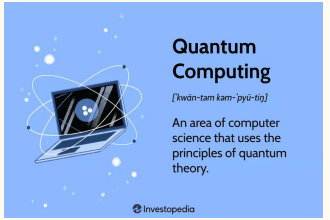


Why Quantum Computing?

- The field of quantum computing is around 5 million dollars
- Has massive capital from both the private and public sectors
- Market valuation by 2035 is expected to reach 116.26 billion dollars
- A report by Mckinsey & Company states that the economic impact and the creation of new industries because of quantum computing will pull in close to 1.3 to 2.7 trillion dollars by 2035.
- Quantum computing is playing a major role in drug discovery, with the timeline for creating a new drug is between 10 to 15 years with costs close to 2 billion dollars.
- Thanks to the advanced algorithms of quantum computing research are able to map molecular orbits and binding affinities for complex proteins in a fraction of the time.



Research Questions

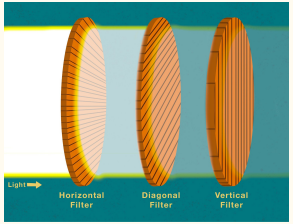


1. How does quantum computation fundamentally differ from classical computation?
2. How can the behavior of qubits be represented mathematically and visualized computationally?
3. What roles do superposition and entanglement play in enabling quantum computation?
4. How do common quantum gates transform quantum states, and how can these transformations be simulated using Qiskit?
5. How do introductory quantum algorithms, such as Grover's Search Algorithm, demonstrate computational advantages over their classical counterparts?
6. What mathematical concepts: including vectors, complex numbers, matrices, and probability amplitudes, form the foundation of quantum computing?
7. What are the current practical limitations of quantum computing hardware, and which challenges must be overcome before large-scale quantum computers become widely available?
8. How can interactive simulations improve conceptual understanding of quantum mechanics and quantum information science?

Motivation for Project

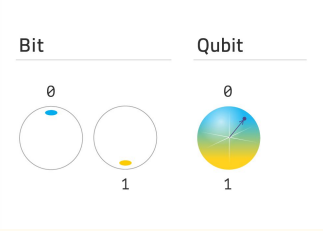
Quantum computing has emerged as one of the most promising interdisciplinary fields at the intersection of physics, mathematics, computer science, and engineering. Unlike classical computers, which process information using binary bits, quantum computers exploit quantum mechanical phenomena such as superposition and entanglement to perform computations in fundamentally different ways. As advances in quantum hardware continue to accelerate, understanding the principles behind quantum computation is becoming increasingly valuable for students interested in computational science and emerging technologies. Ultimately, this introductory project serves as both a personal learning experience and the beginning of a longer journey toward undergraduate research in computational physics and quantum computing. By building simulations from first principles and documenting the underlying mathematics, I hope to establish a strong foundation for future coursework, research opportunities, and continued exploration of quantum information science at the University of Pittsburgh.

Why Does Quantum Computing Matter



- Quantum computing is the process of using concepts like entanglement and superposition to solve complex problems that many traditional computers are unable to solve in a timely manner.
- It helps create a path for more complex problems to be solved, it covers areas such as optimization, machine learning and sampling and simulation.
- Some of the advantages in quantum computing can be seen in a couple of topics. For example, in optimization, you can use complex challenges to optimize decisions which exist everywhere.
- In sampling and simulation, it can be used in physics simulations and enhancing modeling. Finally, in machine learning, quantum computing is able to advance the capabilities of machine learning with more model training for better results.
- Some of the limitations of quantum computing can be seen in cybersecurity, as modern day computers rely on complex problems which take years to solve, quantum computers are able to pose a threat to these standards leaving many modern day security standards exposed.
- This technology is very exciting and interesting as it's able to harness the power of quantum mechanics to solve data faster than regular computers.

Quantum Computing Basics

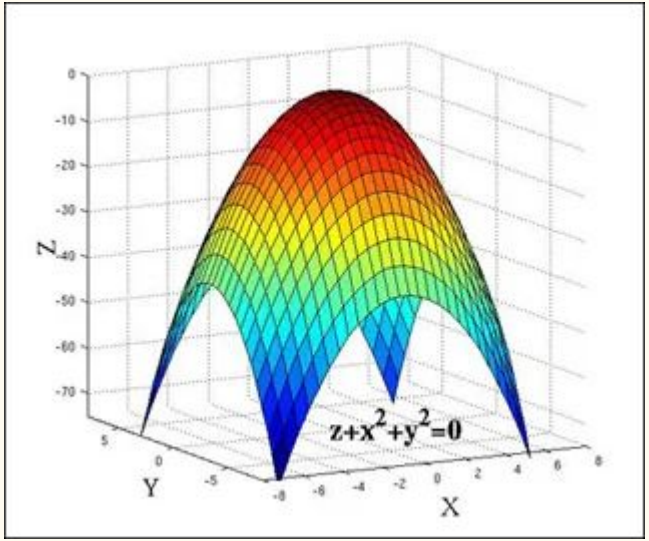


- A bit in quantum computing is referred to as a qubit and it's often referred to as the basic unit of information or data within the quantum computing sector. This is similar to a traditional bit which is used in everyday computers.
- These small units of information within quantum computing are being used to create and measure many quantum particles which are known as photons, electrons or atoms and trapped ions. Qubits are also able to store a much larger amount of information than the traditional bit which is found within computers.
- Furthermore, these units of data are able to perform much more advanced computations and mathematical procedures which would take the regular computer a much longer time.
- When many scientists encounter a difficult problem, they are forced to look towards supercomputers as they have some problems such as how to model the individual atoms in a molecule, that you cannot solve with a regular computer.

Mathematics Behind Quantum

Some of the main mathematical concepts related to quantum computing are listed below, these are topics which make up the skull of quantum computing and what it's going to be within the coming years:

- Vectors
- Complex Numbers
- Matrices
- Linear transformations
- Probability amplitudes
- Inner products



Mathematics Behind Quantum Pt.2

A vector as explained in multivariable calculus is a representation of a localized geometric entity. A vector can be represented and is often represented as either position, velocity or acceleration. However within quantum computing, a physical system is defined as a state vector which lives within a more complex vector space known as the Hilbert Space. The standard basis vectors of i and j are able to map the basic states of 0 and 1. Vectors in calculus are graphed on the cartesian plane, whereas a qubit vector is mapped out on a Bloch Sphere written as:

$$|\psi> = cos(\frac{\theta}{2})|0> + e^{i\phi}sin(\frac{\theta}{2})|1>$$

Mathematics Behind Quantum Pt.3

A Hilbert Space acts as a generalization of Euclidean space with dimensions which are much more extended. The concept was originally introduced by David Hilbert when he was figuring out quadratic forms with an infinite amount of variables. A Hilbert Space is defined as an inner product space. A complex vector space which is used for specific mapping techniques referred to as the inner product. For something to be considered as an inner product space, it must satisfy the criteria listed below:

- Conjugate Symmetry: $< x, y > \ = \ < y, x >$
- Linearity: $< \alpha x + \beta y, z > \ = \alpha < x, z > \ + \ \beta < y, z >$
- Positive Defineness: $< x, x > \ \geq \ 0 \ and \ < x, x > \ = \ 0, \ implies \ that \ x \ = 0$

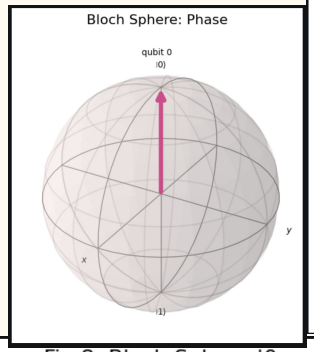
The Bloch Sphere

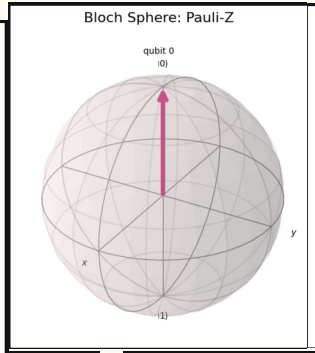
- I like to think of a Bloch Sphere very much like a unit circle in trigonometry.
- Each angle and degree on the unit circle is connected to many of the mathematics in trigonometry and beyond.
- Calculus also utilizes a lot of the unit circle when working with the six trig functions.
- The next slide has some 3D visualizations created using Google CoLab and Claude.

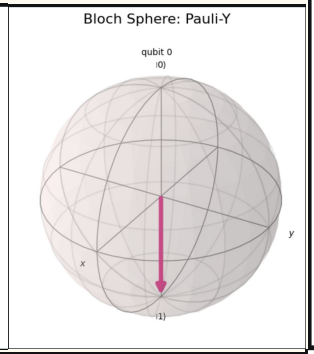
The Unit Circle/Bloch Sphere:

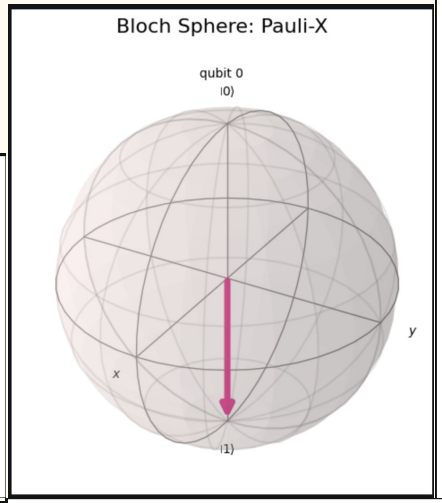
- 2D Circle/Bloch Sphere
- Equation: $x^2 + y^2 = 1$ / $x^2 + y^2 + z^2 = 1$
- Uses real numbers / Uses complex numbers
- Represents trig angles / Represents qubit probabilities and phases

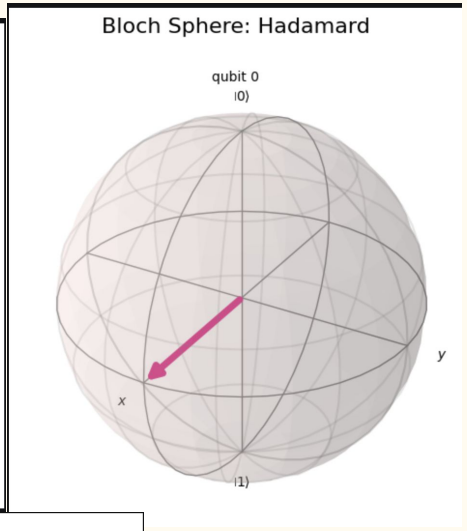
Bloch Sphere Pt.2

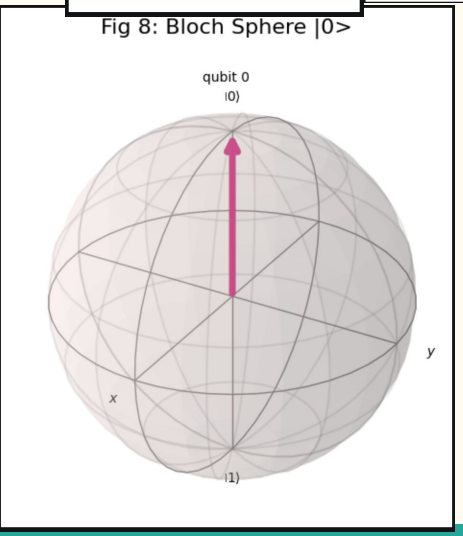


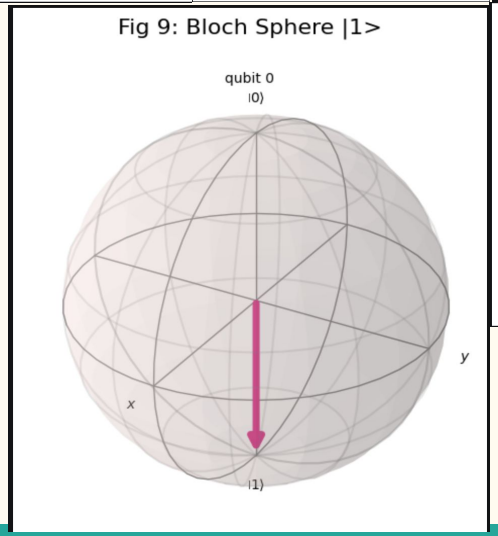


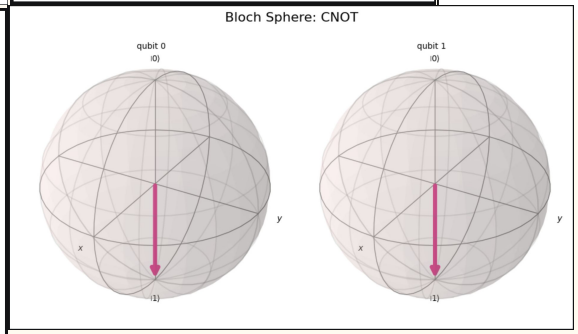


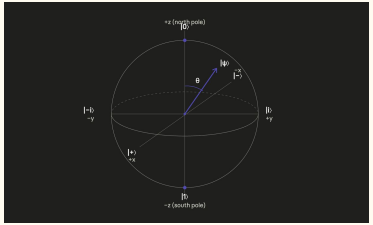


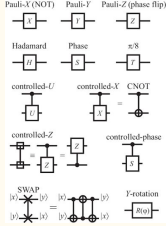










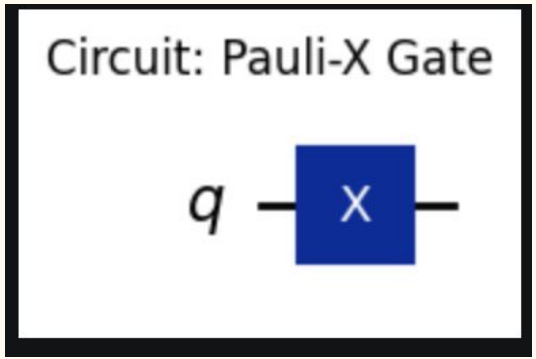


Quantum Gates

- In my classical computing scenarios, there are the three main logic gates referred to as AND, OR, NOT and the standard bits of 0 and 1 which are used to perform many standard arithmetic operations. In quantum computing, a quantum gate is an intervention which manipulates the state of more than one qubit.
- Key Facts:
- A quantum gate is considered a unitary matrix while making sure it follows the two main laws of quantum mechanics:
- Conservation of Probability: Total probability of all measurement outcomes, a quantum gate will function similar to a Bloch Sphere as it's able to change the orientation of a state vector without altering its length.
- Reversibility: A quantum logic operation is also reversible meaning that running a quantum state through a gate, you are able to undo the calculation.

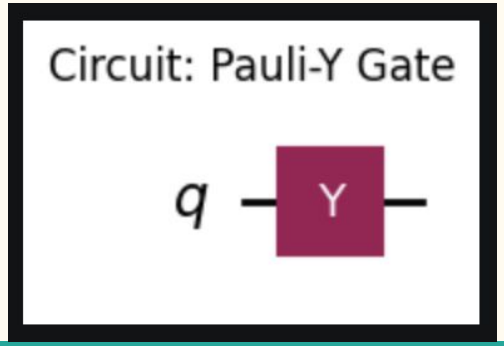
X-Gate

- Mainly used for bit-flipping data, initializes a qubit from the base state to the regular state, used for constructing arithmetic operations like quantum half-adders. Major component for creating control flow conditions within quantum algorithms.
- Sends the Bloch Sphere to the amplitudes of 1, considered the South Pole. Is the quantum NOT gate flips both 0 and 1.
- Around a 180 degree rotation on the Bloch Sphere from the x-axis.



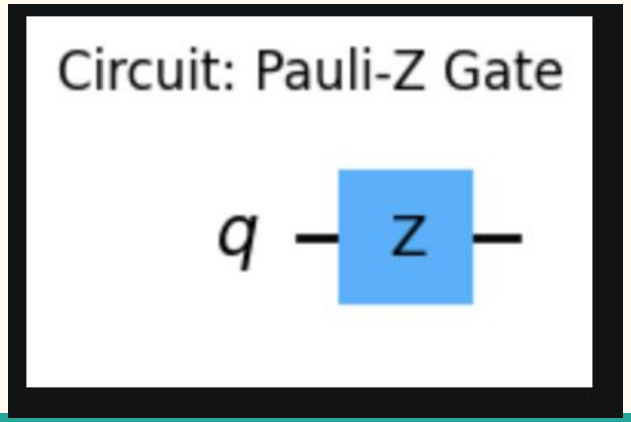
Y-Gate

- Used in bit-flipping and a phase flip, used heavily in quantum error correction codes, used for detecting phase and bit errors, helps map molecular orbitals within quantum chemistry simulations.
- On the Bloch Sphere, it's a 180 degree rotations towards the y-axis
- In the field of quantum chemistry, when a molecular Hamiltonian, which represents the total energy of a molecule is mapped out into qubits, it results in Pauli Strings.
- In a circuit, it's shown as a square with a Y connected to a qubit wire.



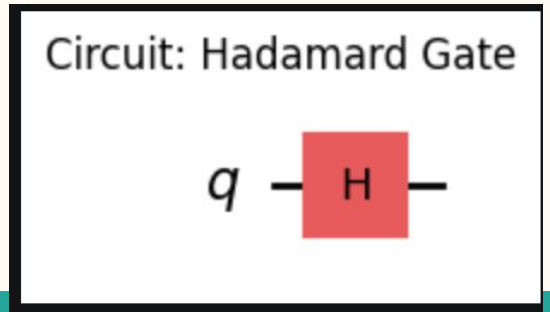
Z-Gate

- Flips the sign when the phase component is within the superposition state without changing the measurement probabilities. Used very heavily for the Grover's Search Algorithm.
- Leaves to 0 amplitude alone and multiplies the 1 amplitude with -1. The invisible measurement but real phase is what really makes interference work.
- On the Bloch Sphere it's a 180 degree rotation towards the z-axis and it's eigenstates are 0 and 1.



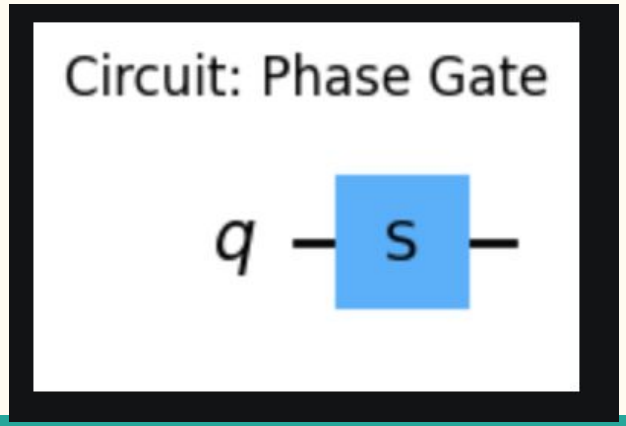
Hadamard Gate

- Represents the 4 position matrix of 3 single digit ones and one negative one times the fraction of 1 over the square root of 2. The circuit on the other hand is a square box with an H attached to a qubit wire. This quantum gate is fundamental towards the exploration on quantum superposition
- Sends the eigenvalues of 0 and 1 to the + and - eigenvalues, creates superposition from a definite state, also applied in reverse it collapses a superposition back down to the computational basis.
- It's a 180 degree rotation from the diagonal axis and its own inverse.



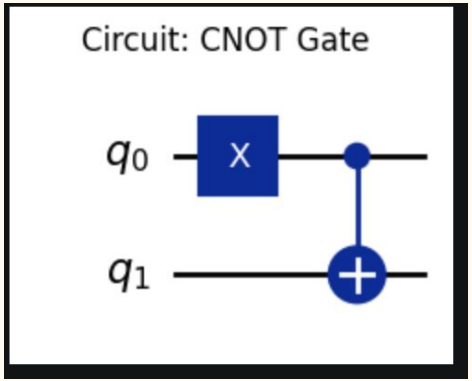
Phase Gate

- The 4 way matrix representation is of a 1, 0, 0 and i along with being equal to the e of the power to i(3.14) / 2. The circuit is a square box with either an S or P on a qubit wire. This gate rotates the state vector by 90 degrees around the Z-axis of the Bloch Sphere, behaving like the square root of the Z gate.
- Is a core component of the Quantum Fourier Transform, is the main computational engine behind the fast prime factorization in cryptanalysis.



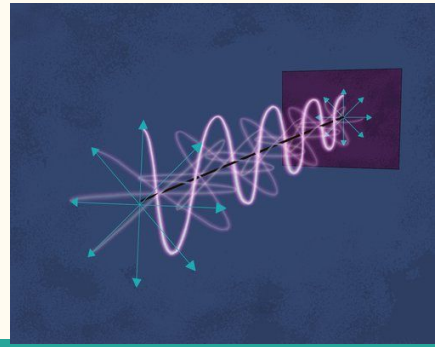
CNOT Gate

- The matrix is represented on a 2 qubit system within a 4 dimensional basis. The circuit is also much more complex in comparison to the other quantum gates as it spans across two parallel qubit wires, a solid control dot is able to sit on the control qubit wire which is then connected to a target node on a qubit wire.
- This gate is also fundamental because when it's combined with the Hadamard gate, the CNOT is a major tool to help generate quantum entanglement which leads to the creation of Bell States.



Superposition

- Superposition is like touching the surface of a body of water at two different points at the same time. This would cause waves to spread out and eventually overlap each other to create a more complex pattern in the water itself.
- This is called superposition, a topic in quantum computing where electrons and photons have many wavelike properties which combine and become superposed.
- Quantum waves unlike the one formed on a body of water are mathematical as they are expressed on various equations which are used to help describe a certain outcome or probability of an object.



$$|\alpha|^2 + |\beta|^2 = 1$$

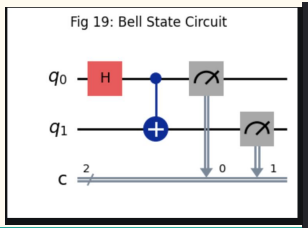
Quantum Measurement

- Checking the value of a bit in classical computing, does not cause any shockwaves, whereas in quantum computing, checking the value of a qubit forces it to become a traditional bit causing high levels of disruption.
- Before any measurement, a qubit exists in a state called the wave function, the function describes the evolution of probabilities where the qubit is both 0 and 1.
- After a classical device interacts with the qubit, the wave function collapses which forces the state vector to snap in a computational axes of 0 or 1. Once the wave function collapses all previous quantum information is destroyed.
- Quantum mechanisms are all probabilistic, therefore it is hard to predict with absolute certainty what a single qubit will do upon being measured.

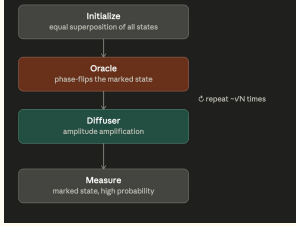
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Entanglement

- Quantum entanglement is when a pair of particles are generated, share spatial proximity in a way that the quantum state cannot be described with the others.
- The state of one entangled particle is know dictating the state of it's partner, both particles will now exist as one quantum system.
- To be able to entangle two qubits, we must put them into a state that cannot be factored into two independent qubits. This is called a Bell State
- There are four bell states and the most famous one along with one which is most commonly used is the density operator squared state. Neither qubit has a definite state either qubit can have an amplitude of 0 or 1.

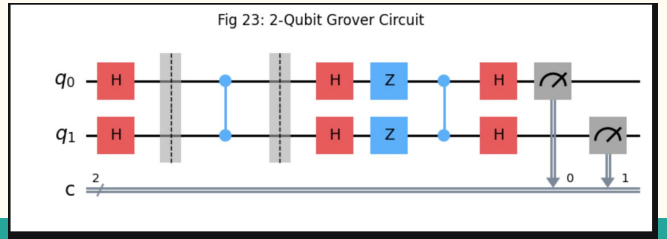


$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



Grover Search

- Amplitude amplification is the main engine running behind the Grover Search algorithm.
- After initialization every one of the N possibilities has an equal, tiny amplitude.
- The oracle flips the sign of the target's amplitude (using the Z-type phase trick from earlier
- The diffuser then reflects all amplitudes about their average, which geometrically rotates the whole state a little toward the marked answer.
- Each oracle+diffuser round nudges probability onto the target, and after about $\sqrt{N}$ rounds it's nearly certain, versus the $\sim N/2$ checks a classical search needs.



Real Applications

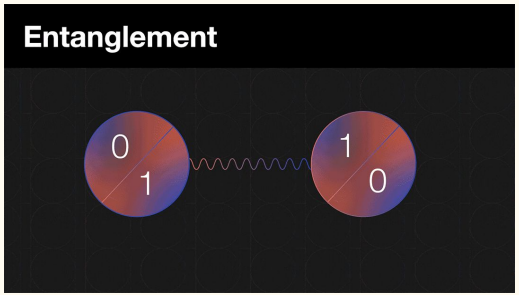
- Optimization and finance: portfolio balancing, routing and scheduling lean on quantum annealing and Quantum Approximation Optimization Algorithm.
- Drug discovery and materials are a part of quantum chemistry, along with simulating molecules and reactions using a Variational Quantum Eigensolver.
- Quantum AI and machine learning explore quantum kernels and variational classifiers.

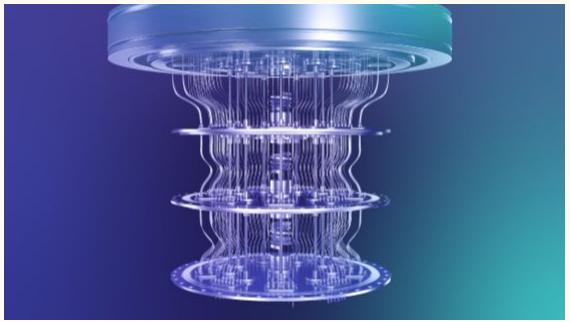
The main problem is when there are problems where state space is growing exponentially, which is where superposition and entanglement might be a advantage.

Limitations

- The main problem within the field of quantum computing is decoherence as qubits are losing their quantum state through noise from the environment in a matter of seconds.
- Many gate operations like CNOT are showing lots of errors.
- Machines are also NISQ with dozens to many hundreds of physical qubits having error rates too high for long algorithms.
- Quantum error correction fixes NISQ but is very expensive as it can take thousands of physical qubits to protect one reliable qubit.

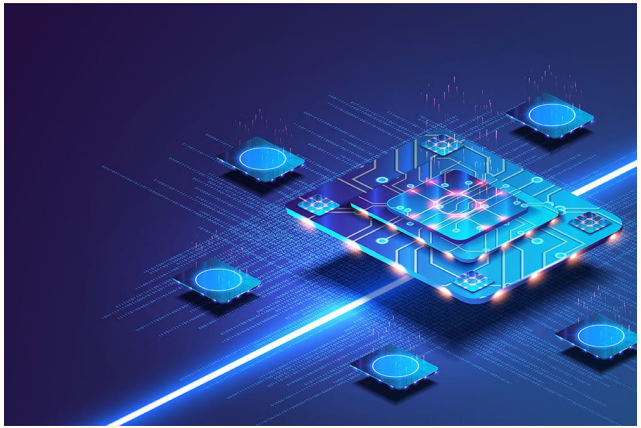


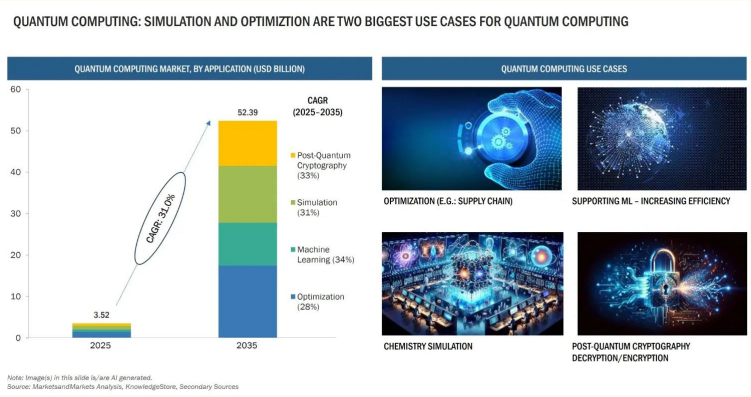




Future of Quantum Computing

- IBM and Google are building superconducting qubits
- IonQ and Quantinuum are using trapped ions
- Microsoft is pursuing topological qubits, also providing for the Azure Quantum Software Layer.





Project Summary/Future Work

- Quantum Computing Playground is an independent computational physics project exploring the mathematical foundations and computational principles of quantum computing through interactive simulations built with Python and Qiskit.
- The project investigates how quantum systems fundamentally differ from classical computers by modeling qubits, quantum gates, superposition, entanglement, and introductory quantum algorithms. Rather than focusing solely on programming, this repository emphasizes the underlying mathematics, physics, and visualization of quantum phenomena through scientific computing.
- Some of my future work contains a European colonialism data driven analysis project involving Google Colab and various GIS software technique, thank you guys so much for being a part of this journey and stay tuned.

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